

TDDD41-Lab3

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1 Clustering and Association Analysis on the MONK-1 Dataset

1.1 Introduction

The objective of this exercise is to investigate whether clustering algorithms are able to recover the class structure of the MONK-1 dataset and to understand the importance of the distance metric used in clustering. The MONK-1 dataset contains 124 instances described by six nominal attributes and a binary class label.

1.2 Clustering Experiments

clustering algorithms were applied using “Classes to clusters evaluation” mode it was used to compare clusters with the true class labels.

1.2.1 K-Means Clustering

K-Means clustering was performed with different numbers of clusters.

K = 2 Incorrectly clustered instances: 59 (47.58%)

K = 3 Incorrectly clustered instances: 70 (56.45%)

K = 4 Incorrectly clustered instances: 76 (61.29%)

K = 5 Incorrectly clustered instances: 79 (63.71%)

Although the within-cluster sum of squared errors decreased as the number of clusters increased, the misclassification rate increased. Even with $K = 2$, which matches the number of classes, the clustering did not correspond to the true class division.

1.2.2 EM Clustering

The EM algorithm was applied with automatic selection of the number of clusters using cross-validation. The algorithm selected only one cluster.

Incorrectly clustered instances: 62 (50%)

This corresponds to random assignment since the dataset is balanced (62 instances per class). The EM algorithm did not detect any natural probabilistic separation in the data.

1.2.3 Hierarchical Clustering

Hierarchical clustering with single linkage and $K = 2$ produced:

Incorrectly clustered instances: 61 (49.19%)

Almost all instances were placed into one large cluster due to the chaining effect of single linkage.

1.3 Why Clustering Cannot Recover the Class Structure

All clustering algorithms failed to discover the class division. The reason is that clustering algorithms rely on distance measures (e.g., Euclidean distance) and assume that instances belonging to the same class form compact and well-separated regions in the attribute space.

However, in the MONK-1 dataset, the class is defined by the logical rule:

$$(A5 = 1) \vee (A1 = A2)$$

This logical rule creates multiple disconnected regions for class 1 rather than a single compact cluster. Therefore, instances of class 1 are spread across the attribute space and do not form a geometrically coherent group.

Since distance-based clustering algorithms optimize within-cluster similarity rather than logical consistency, they cannot recover this class structure.

1.4 Association Rule Mining

Association analysis was performed using the Apriori algorithm with the following parameters:

- Minimum support: 0.05
- Minimum confidence: 0.9
- Maximum number of rules: 19
- Class attribute as the target

The following high-confidence rules were obtained:

- $\text{attribute\#5} = 1 \Rightarrow \text{class} = 1$

- $\text{attribute\#1} = 1 \wedge \text{attribute\#2} = 1 \Rightarrow \text{class} = 1$
- $\text{attribute\#1} = 2 \wedge \text{attribute\#2} = 2 \Rightarrow \text{class} = 1$
- $\text{attribute\#1} = 3 \wedge \text{attribute\#2} = 3 \Rightarrow \text{class} = 1$

Each rule has confidence 1. These four rules perfectly describe class 1.
The rules can be summarized as:

$$(A5 = 1) \vee (A1 = A2)$$

Thus, association analysis successfully uncovered the underlying logical definition of the class.

1.5 Discussion

Clustering algorithms did not perform poorly; rather, they optimized a different objective function. K-Means minimized within-cluster variance, EM selected the statistically optimal number of clusters, and hierarchical clustering followed its linkage criterion.

The failure to match the class labels arises because the true class structure is logical rather than geometric. Distance-based clustering assumes compact and well-separated clusters, which is not the case for MONK-1.

1.6 Conclusion

This experiment demonstrates that clusters found by distance-based algorithms do not necessarily correspond to human-defined class structures. The choice of distance metric and clustering objective is crucial. In datasets where classes are defined by logical rules rather than geometric proximity, clustering algorithms may not recover the true class division, while association analysis can successfully identify the underlying structure.